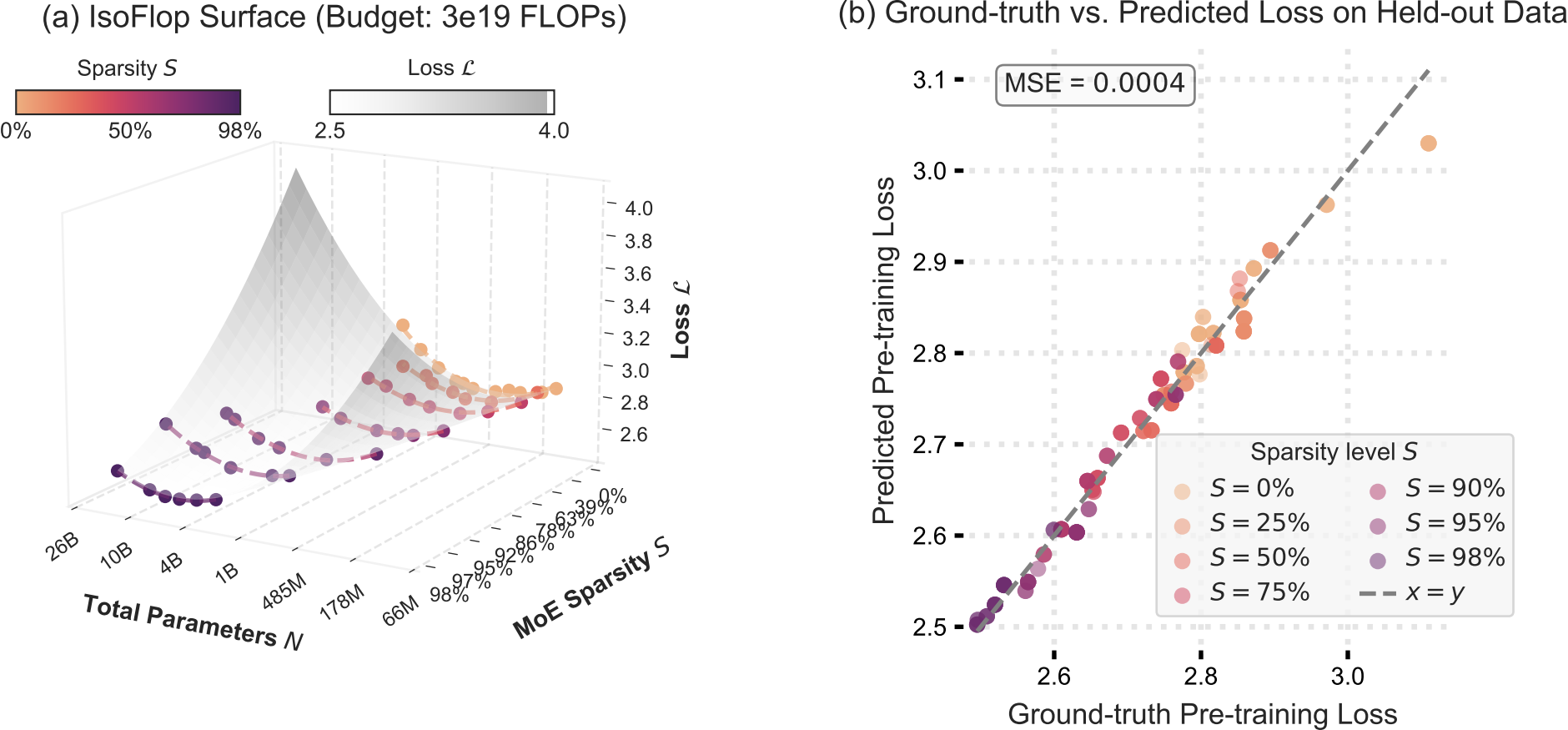
(a) IsoFlop Surface (Budget: 3e19 FLOPs)
Sparsity S
0%
50%
98%
Loss $\mathcal{L}$
2.5
4.0
4.0
3.8
3.6
3.4
3.2
3.0
2.8
2.6
Loss $\mathcal{L}$
26B
10B
4B
1B
485M
178M
66M
Total Parameters N
0%
39%
63%
78%
86%
92%
95%
97%
98%
MoE Sparsity S
(b) Ground-truth vs. Predicted Loss on Held-out Data
MSE = 0.0004
3.1
3.0
2.9
2.8
2.7
2.6
2.5
Predicted Pre-training Loss
2.6
2.8
3.0
Ground-truth Pre-training Loss
Sparsity level S
$S = 0\%$
$S = 25\%$
$S = 50\%$
$S = 75\%$
$S = 90\%$
$S = 95\%$
$S = 98\%$
$x = y$